PHASE 2

INN SYNC

Phase 2 Engineering Handoff

Guest App · Travel Itinerary

- Where we are — HMS Core progress
- How HMS works as your foundation
- What you are building — Guest App
- What you are building — Travel Itinerary
- How to build it right

Inn Sync · Kigali · 2026

ABOUT INN-SYNC

What we are building — and why it matters

Inn-Sync is Rwanda's first intelligent hospitality and Tourism platform — built to replace spreadsheets, WhatsApp and guesswork with a single connected system that runs every hotel, delights every guest, plans every trip and tracks every carbon footprint.

No native HMS exists

Every Rwandan hotel runs on spreadsheets and WhatsApp. Revenue leaks daily. Guests experience a 2005-era stay in a 2026 country.

Rwanda tourism is accelerating

$498M in tourism revenue. 1,500+ properties. KCC summits, gorilla season, MICE growth. Hotels need infrastructure to keep up.

We are the team to build it

Built by people who understand Kigali, Kinyarwanda, MoMo, RDB and Volcanoes NP. Local knowledge is our permanent advantage.

THE PLATFORM

Four pillars. One shared foundation.

01 HMS Core

LIVE — Phase 1

The operational backbone. Reservations, housekeeping, front desk, billing, staff, analytics. Your foundation API.

BUILT BY TEAM 1

02 Guest mobile app

OUR MISSION

Guest-facing experience layer. Digital check-in, BLE key, AI concierge, F&B ordering, express checkout.

TK - Lead Dev

03 Travel itinerary

OUR MISSION

AI-powered Rwanda trip planner. Gorilla permits, park routes, transport, cross-border EAC journeys.

TK - Lead Dev

04 Green certification

Phase 3

IoT energy tracking, per-stay carbon footprint, eco guest scoring, Green Globe auto-certification.

Not Decided

HMS CORE — WHAT TEAM 1 IS DELIVERING

The foundation our work is built on

IN PRODUCTION

Reservations & PMS

Booking engine · room inventory · availability calendar · OTA sync · group bookings

Front desk

Check-in/out · room rack · walk-ins · early/late requests · room assignment

Housekeeping

Task assignment · room status · linen tracking · inspection logs · mobile for staff

Billing & payments

Folios · MoMo / card / cash · multi-currency RWF · RRA tax · split billing

Staff management

Roles · shifts · RBAC permissions · task tracking · incident logging

Reporting & analytics

RevPAR · TRevPAR · occupancy · RDB compliance reports · channel analysis

HMS CORE — TECHNICAL ARCHITECTURE

How it is built — so you can build on top of it

Frontend layer
Next.js + React

- Role-based staff dashboard
- Receptionist · manager · housekeeper views
- Mobile-responsive for tablet at front desk
- Real-time room updates via WebSockets
- Offline mode with local sync

API layer
Node.js REST + GraphQL

- JWT authentication + RBAC middleware
- RESTful endpoints for all HMS operations
- GraphQL for complex dashboard queries
- Webhook system for third-party integrations
- Rate limiting + request validation

Data layer
PostgreSQL + Redis

- PostgreSQL — guests, reservations, billing
- Redis — session cache, real-time room state
- Background job queue for reports & emails
- Daily encrypted backups
- Rwanda data residency compliant

The HMS API should be the single source of truth · Our guest app reads from it · Never duplicate data · Always call the API

HMS CORE — THE API CONTRACT

What the HMS exposes to our applications

Reservations API

- GET /reservations/:id — fetch booking detail
- POST /reservations — create new booking
- GET /rooms/availability — real-time inventory
- PUT /rooms/:id/status — update room state

Guests API

- GET /guests/:id — full guest profile
- POST /guests — create profile at booking
- PUT /guests/:id/preferences — update prefs
- GET /guests/:id/history — stay history

Billing API

- GET /folios/:reservation_id — guest folio
- POST /folios/:id/charge — add charge
- POST /payments — process MoMo / card
- GET /invoices/:id/pdf — generate receipt

Operations API

- GET /housekeeping/tasks — today's task list
- PUT /tasks/:id/status — mark complete
- POST /service-requests — guest request
- GET /staff/schedule — on-duty roster

The problems Phase 2 exists to solve

Guests queue at front desk for 15 minutes on arrival

→ Online check-in + BLE digital key. Walk straight to the room.

Guest app

No room key = no arrival experience. Cards get demagnetised.

→ Phone is the key. Bluetooth unlock. No card. No queue. No failure.

Guest app

Room service ordered by phone — lost, forgotten, slow

→ In-app F&B ordering. Direct to kitchen. Tracked. Timed.

Guest app

Guests plan Rwanda trips through 4 different WhatsApp threads

→ One AI-generated itinerary: parks, permits, transport, all booked in-app.

Itinerary

Gorilla permits booked separately — guests miss out or overpay

→ Real-time RDB permit availability inside the app. One tap to confirm.

Itinerary

Hotels lose $2,000+/month in off-system tour commissions

→ All activities booked through the platform. Revenue captured, tracked.

Both

GUEST MOBILE APP — FULL OVERVIEW

What you will build and what it does

OUR MISSION

Pre-arrival

- Online check-in form
- Passport / ID upload
- Room preference selection
- Carbon offset opt-in
- Advance payment settlement
- First upsell — room upgrade

Arrival

- Room ready push notification
- Digital key via Bluetooth
- Bypass front desk entirely
- Day 1 itinerary surfaced
- F&B pre-order option
- AI concierge introduction

In-stay

- AI concierge 24/7 chat
- Room service ordering
- Housekeeping requests
- Spa & tour booking
- Maintenance reporting
- Upsells at optimal moments

Departure

- Folio review in-app
- Bill dispute flow
- MoMo / card settlement
- Express check-out button
- Eco score badge issued
- Review prompt + loyalty points

GUEST APP — AI CONCIERGE

How the intelligent concierge works end to end

Guest types message — App sends to backend — Backend calls LLM + tools — LLM decides & acts

Function tools available to the LLM:

Get Booking Details (guest_id)

→ Returns stay dates, room, hotel info

Get Room Status (room_id)

→ Returns clean/ready/occupied status

Check Permit Availability (park, date)

→ Returns available permit slots + price

Place Room Service Order (items, time)

→ Posts to HMS, returns order confirmation

Book Service Request (type, detail)

→ Creates housekeeping / maintenance task in HMS

Get Itinerary Suggestions (guest id, date)

→ Returns AI-generated activity list from itinerary

Guest: "Can I still get a gorilla trek permit for Thursday?"

AI: "Thursday has 2 permits remaining for the Susa group at Volcanoes NP — $1,500 per person. Departure is 5:30am. I can book for you and add an early breakfast at 4:45am. Shall I proceed?"

TRAVEL ITINERARY — FULL OVERVIEW

Turning Rwanda into one seamless journey

OUR MISSION

Experience & content database

Structured data on every Rwanda tourism experience — parks, tours, F&B, culture, transport. Geo-tagged, priced, seasonality-flagged. The AI cooks with this data. Quality here = quality everywhere.

Wildlife permit hub

Real-time RDB API integration for Volcanoes, Nyungwe and Akagera. Gorilla permits ($1,500/person), chimp permits, game drives — all bookable inside the app with live availability. This is Inn Sync's biggest competitive moat.

AI itinerary generator

Guest profile from HMS + content database ⟶ LLM generates a full sequenced day-by-day itinerary. Accounts for drive times (Volcanoes = 3hrs from Kigali), permit availability, guest energy levels, budget tier.

Transport & logistics

Vetted 4WD operators, domestic flights (RwandAir Kigali→Akagera), Lake Kivu boats, airport transfers. All bookable in-app. Every booking generates a carbon record fed to the Green Cert engine.

Live replanning engine

Permit closed? Road blocked? Rain forecast on canopy walk morning? System detects via API monitoring + weather data and proposes an alternative itinerary with one-tap guest confirmation.

Regional gateway (EAC+)

Extend itineraries into Uganda (Bwindi gorillas), Tanzania (Serengeti via RwandAir) and DRC. Visa rules, border crossing times, partner operator bookings. No competitor plans multi-country East African trips natively.

TRAVEL ITINERARY — BUILD SEQUENCE

The exact order to build the itinerary module

01

Content database (build first)

Before any AI, build the structured experience database. Schema: experiences, operators, locations, availability_slots, pricing_tiers. Admin panel for content team to add/edit. Every other feature depends on data quality here.

02

RDB permit API integration

Contact Rwanda Development Board immediately — API access has a 2-3 month lead time. Build against a mock API you control while waiting. Endpoints needed: permit availability check, permit hold, permit confirm, permit cancel.

03

Transport operator integrations

Partner with 3-5 vetted 4WD operators and build a simple booking webhook. Standardise the payload so adding new operators is trivial. Include: vehicle type, capacity, price, carbon factor for green cert.

04

AI generator (backend service)

POST /itinerary/generate. Body: guest_id, arrival_date, departure_date, interests[], budget_tier. Backend calls LLM with content DB as tool context. Returns structured JSON: [{day, date, activities[], transport, notes}]. Never do this on the client.

05

Guest app surface

Render JSON itinerary as interactive day-by-day view. Guest can swap activities, confirm bookings, see permit status. Connect to HMS folio for billing when activities are booked.

INTEGRATION ARCHITECTURE

How your pillars connect to the HMS — and to each other

HMS Core

Source of truth

- Guest profiles
- Reservations
- Room status
- Billing / folios
- Staff tasks
- RDB reports

Guest app

Reads + writes HMS via API

Reads:	Check-in status, folio, room
Writes:	Guest prefs, service requests
Reads:	Permit availability (itinerary)
Writes:	Payment settlements
Reads:	AI concierge data
Writes:	Review + checkout event

Travel itinerary

Reads HMS · writes to guest app

Reads:	Stay dates + guest profile
Reads:	Room type, budget tier
Calls:	RDB permit API (external)
Writes:	Itinerary to guest app
Writes:	Bookings ⟶ HMS folio
Sends:	Carbon data ⟶ Green Cert

SUCCESS METRICS

How we know Phase 2 is working

<60s
Check-in time
From app open to room door unlocked

4.9/5
Guest satisfaction
In-app rating target post-stay

>70%
App adoption rate
Guests who use app vs front desk

$75+
Ancillary per guest
TRevPAR uplift vs room-only spend

0
Double bookings
Permit + room sync accuracy

<3s
API response time
All HMS calls from app

You are building the part guests see.

Team 1 built the engine. You build the experience. Between the HMS, the guest app and the travel itinerary, Inn Sync becomes the platform that makes Rwanda's hospitality extraordinary — for every guest who arrives, from anywhere in the world.

HMS is your backbone
Call its APIs. Trust its data.

Contact RDB this week
Permits are the product's biggest win.

Build in order
Auth ⟶ booking ⟶ key ⟶ services ⟶ AI.

INN-SYNC

Phase 2 — Guest App & Travel Itinerary

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